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1: function EULER_ADAPTIVE( $f, [t_0, t_f], x_0, E_s$ )
2:    $t \leftarrow [t_0], \quad x \leftarrow [x_0], \quad h \leftarrow (t_f - t_0)/100$ 
3:   while  $t_{\text{end}} < t_f$  do
4:      $h \leftarrow \min(h, t_f - t_{\text{end}})$ 
5:      $x_{\text{full}} \leftarrow x_{\text{end}} + h f(t_{\text{end}}, x_{\text{end}})$  ▷ or RK4
6:      $x_{\text{mid}} \leftarrow x_{\text{end}} + \frac{h}{2} f(t_{\text{end}}, x_{\text{end}})$ 
7:      $x_{\text{half}} \leftarrow x_{\text{mid}} + \frac{h}{2} f(t_{\text{end}} + \frac{h}{2}, x_{\text{mid}})$ 
8:      $E_a \leftarrow |x_{\text{half}} - x_{\text{full}}|$ 
9:     if  $E_a \leq E_s$  then
10:       append  $t_{\text{end}} + h$  to  $t$ 
11:       append  $x_{\text{half}}$  to  $x$ 
12:     end if
13:      $h \leftarrow 0.9 h \sqrt{E_s/E_a}$  ▷ b/c  $E_a$  is  $O(h^2)$ 
14:   end while
15:   return  $t, x$ 
16: end function

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